

Python Programming - 2301CS404

Lab - 6

Roll No. : 459

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01) WAP to find sum of all the elements in a List.

```
In [4]: list = [1,3,5,7]

sum = 0

for i in list:
    sum += i

print("Sum : " , sum)
```

Sum : 16

02) WAP to find largest element in a List.

```
In [6]: list = [2,4,5]

# print(max(list))

# or

mx = list[0]
for i in list:
    if mx < i:
        mx = i

print(mx)
```

5

03) WAP to find the length of a List.

```
In [8]: list = [1,3,5,6]

count = 0

for i in list:
    count+=1

print("Len : ",count)
```

Len : 4

04) WAP to interchange first and last elements in a list.

```
In [10]: list = [1,3,5,6]

temp = list[0]
list[0] = list[-1]
list[-1] = temp

print(list)
```

[6, 3, 5, 1]

05) WAP to split the List into two parts and append the first part to the end.

```
In [16]: list = [1,3,4,5,6]

p = int(input("Enter Position To Split : "))

left = list[:p+1]
right = list[p+1:]

list = right + left

print(list)
```

Enter Position To Split : 2
[5, 6, 1, 3, 4]

06) WAP to interchange the elements on two positions entered by a user.

```
In [11]: list = [1,3,5,6]

p1 = int(input("Enter Position 1 : "))
p2 = int(input("Enter Position 2 : "))

temp = list[p1]
list[p1] = list[p2]
list[p2] = temp

print(list)
```

Enter Position 1 : 1
Enter Position 2 : 3
[1, 6, 5, 3]

07) WAP to reverse the list entered by user.

```
In [ ]: n = int(input("Enter Length : "))

list = []

for i in range(n):
    list.append(int(input(f"Enter {i+1} Element :")))

list = list[::-1]

print(list)
```

```
Enter Length : 4
Enter 1 Element : 2
Enter 2 Element : 3
Enter 3 Element : 5
Enter 4 Element : 5
[2, 3, 5, 5]
```

08) WAP to print even numbers in a list.

```
In [23]: list = [1,2,3,4]

for i in list:
    if(i % 2 == 0):
        print(i)
```

```
2
4
```

09) WAP to count unique items in a list.

```
In [31]: list = [11,11,4,4,66]

unique = 0

for i in list:
    if(list.count(i) == 1):
        unique += 1

print(unique)
```

```
1
```

10) WAP to copy a list.

```
In [35]: list = [1,2,3,4,5]

list2 = list.copy()

# or

# list2 = list # Refrance Same

print(list2)
```

Out[35]: [1, 2, 3, 4, 5]

11) WAP to print all odd numbers in a given range.

```
In [39]: a = int(input("Enter a : "))
b = int(input("Enter b : "))

for i in range(a,b+1):
    if(i % 2 != 0):
        print(i)
```

Enter a : 5

Enter b : 10

5

7

9

12) WAP to count occurrences of an element in a list.

```
In [46]: list = [11,2,2,3,4,5,55,6,11,11,55,11]

for i in set(list):
    print(f"number {i} : occurrence {list.count(i)}")
```

number 2 : occurrence 2

number 3 : occurrence 1

number 4 : occurrence 1

number 5 : occurrence 1

number 6 : occurrence 1

number 11 : occurrence 4

number 55 : occurrence 2

13) WAP to find second largest number in a list.

```
In [50]: list = [1,2,4,6,7]

# for i in list:
#     if(max(list) != i):
#         list2.append(i)

# print(max(list2))

list.sort() # Must Be Remember Its Not Return List

print(list[-2])
```

6

14) WAP to extract elements with frequency greater than K.

```
In [52]: list = [11,2,2,3,4,5,55,6,11,11,55,11]

k = int(input("Enter K : "))

for i in set(list):
```

```
if(list.count(i) > k):  
    print(f"number {i} : occurrence {list.count(i)}")
```

Enter K : 1
number 2 : occurrence 2
number 11 : occurrence 4
number 55 : occurrence 2

15) WAP to create a list of squared numbers from 0 to 9 with and without using List Comprehension.

```
In [56]: list = []  
  
for i in range(0,10):  
    list.append(i*i)  
  
# or comprehencen  
  
list = [i*i for i in range(0,10)]  
  
print(list)
```

[0, 1, 4, 9, 16, 25, 36, 49, 64, 81]

16) WAP to create a new list (fruit whose name starts with 'b') from the list of fruits given by user.

```
In [58]: n = int(input("Enter Length : "))  
  
list = []  
  
for i in range(n):  
    list.append(input(f"Enter {i+1} Element :"))  
  
fruit = []  
  
for i in list:  
    if(i[0] == "b"):  
        fruit.append(i)  
  
print(fruit)
```

Enter Length : 3
Enter 1 Element : apple
Enter 2 Element : banana
Enter 3 Element : mengo
['banana']

17) WAP to create a list of common elements from given two lists.

```
In [59]: list = [1,2,3,4]  
list2 = [1,3,6,8,7]  
list3 = []  
  
for i in list:  
    if(list2.count(i) != 0):  
        list3.append(i)
```

```
print(list3)
```

```
[1, 3]
```